

Validating seabird behaviour inferred from hidden Markov models to inform conservation

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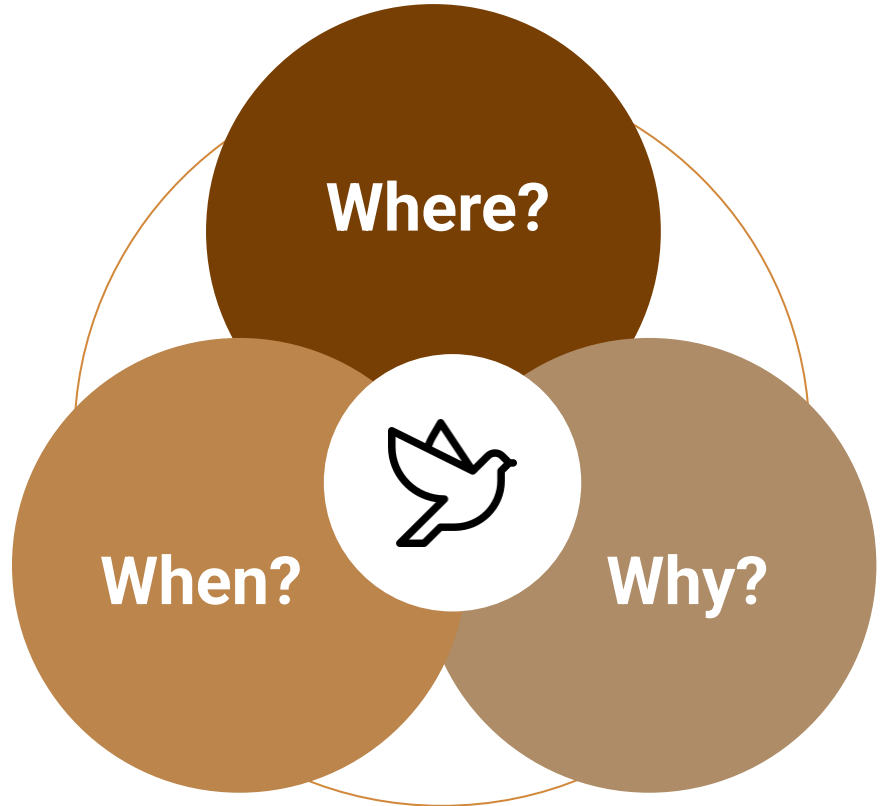


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Background

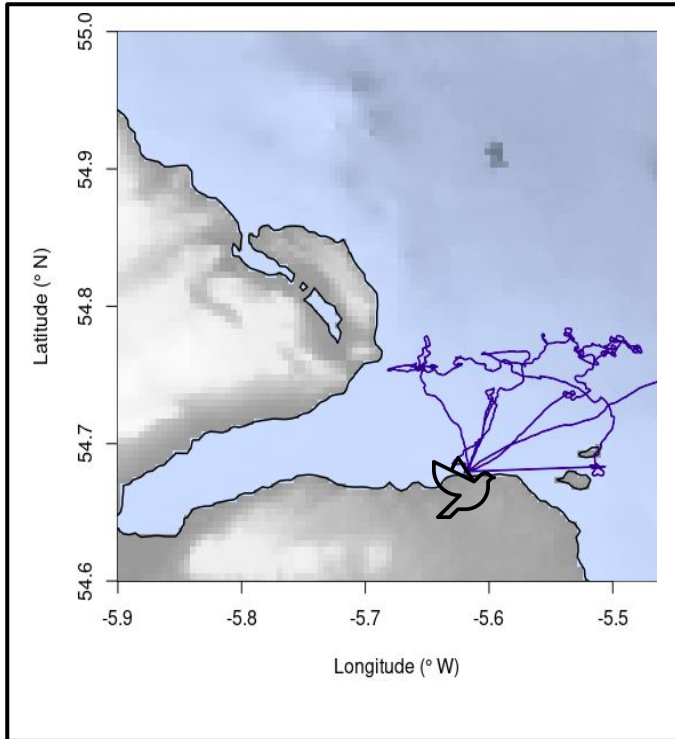


Motivation



- Seabird population decline
- The difficulty to directly observe behaviours influencing movement in the marine environment
- The need to identify at-sea seabird behaviours for conservation actions

Problem



- Seabird behaviours are inferred from telemetry data
- Inferred behaviours are rarely validated
- Observed behavioural data can be difficult to obtain

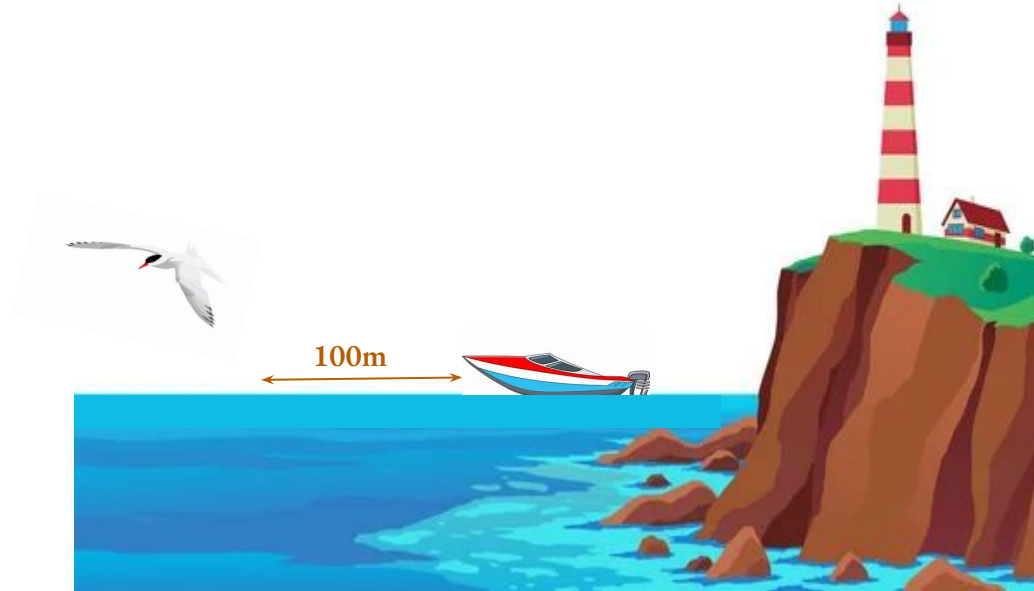
Goal



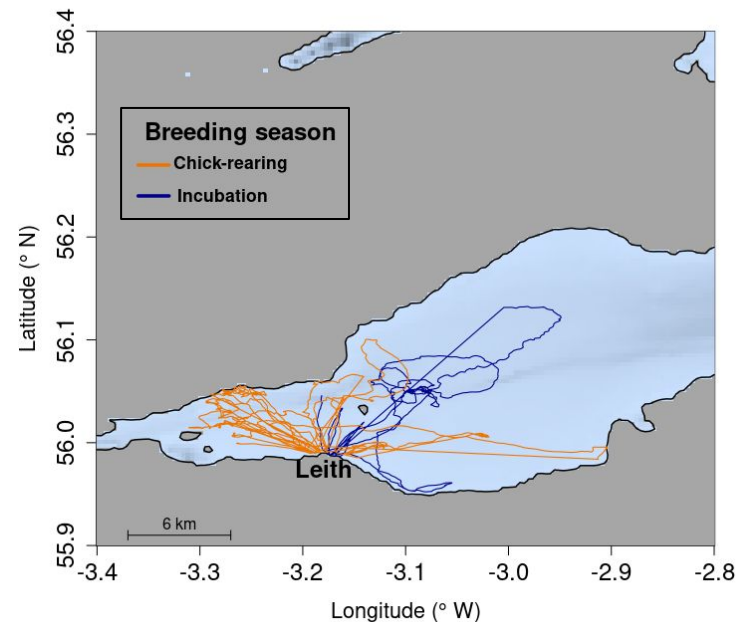
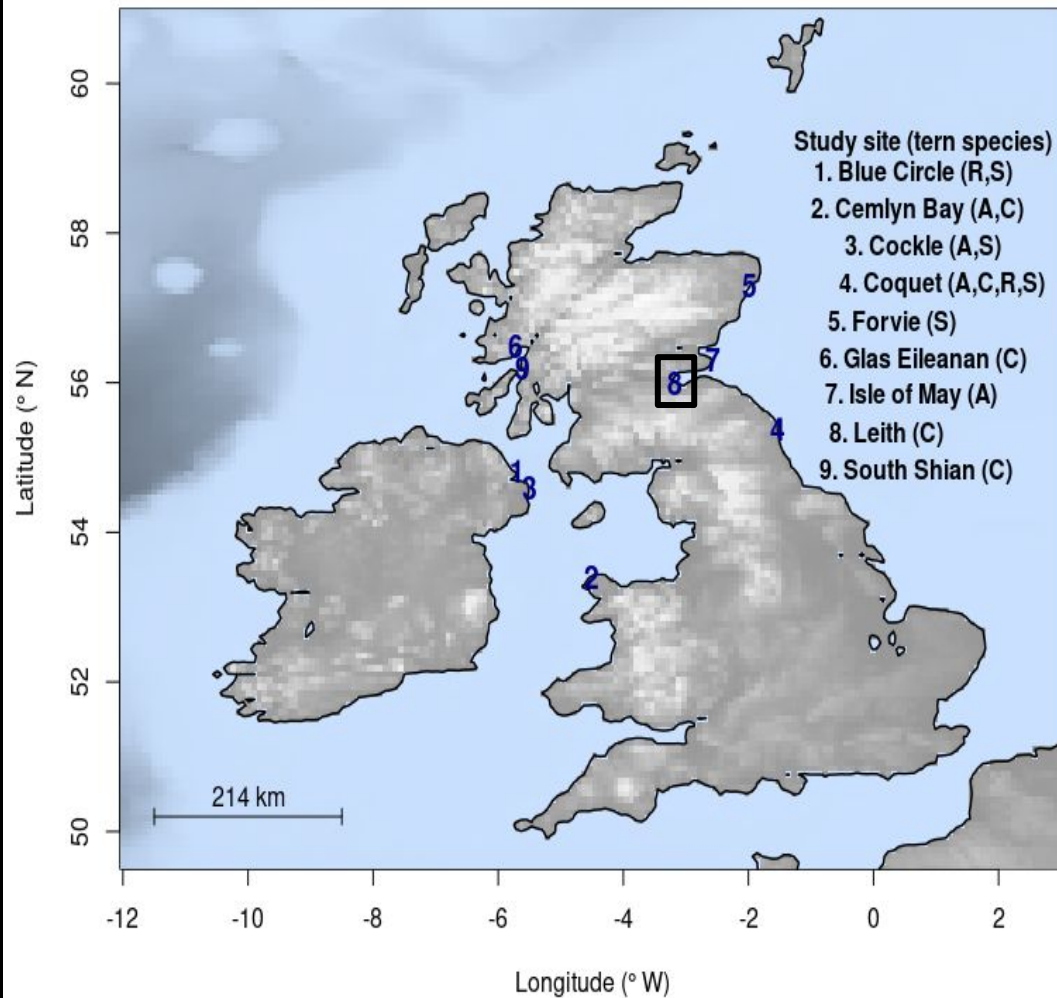
Validate inferred behaviours of seabirds
for effective conservation management

Data

- Collected by : Joint Nature Conservation Committee (JNCC)
- Method : visual tracking¹
- Data : Movement data of boat
(every 1 sec)
+
Behavioural data of terns
(recorded by observers on the boat)
- Period : Incubation + Chick-rearing
- Year : 2009 - 2011



¹Perrow, M. R., Skeate, E. R. & Gilroy, J. J. (2011). Visual tracking from a rigid-hulled inflatable boat to determine foraging movements of breeding terns

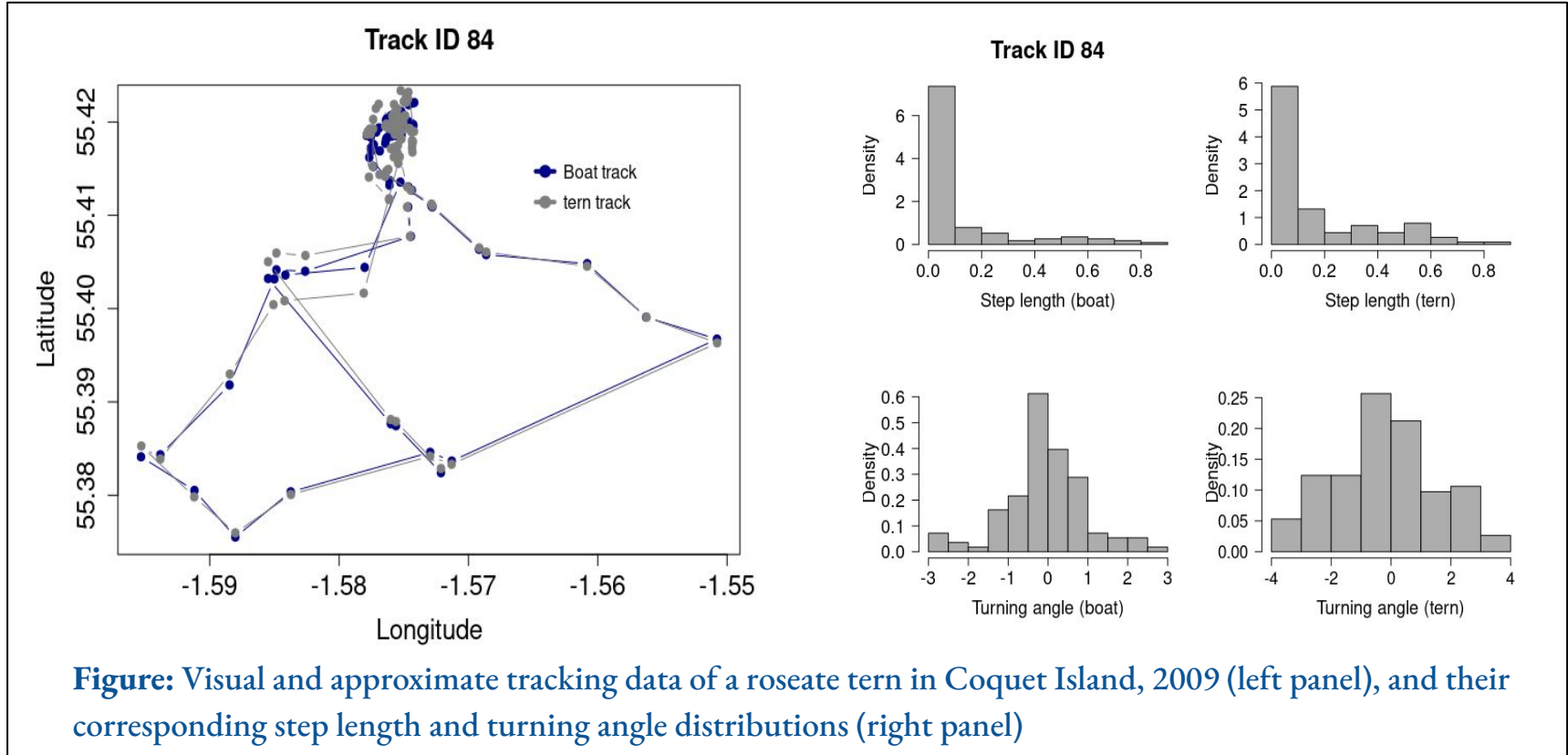


Boat movement data during visual tracking of common terns at Leith, 2010

💡 Work Flow

- 💡 **Assess** boat movement data as a proxy for terns location
- 💡 **Infer** behaviours from visual tracking data using hidden Markov model
- 💡 **Validate** inferred behaviours with observed behavioural data of terns

Assessing Boat Movement Data



Assessing Boat Movement Data

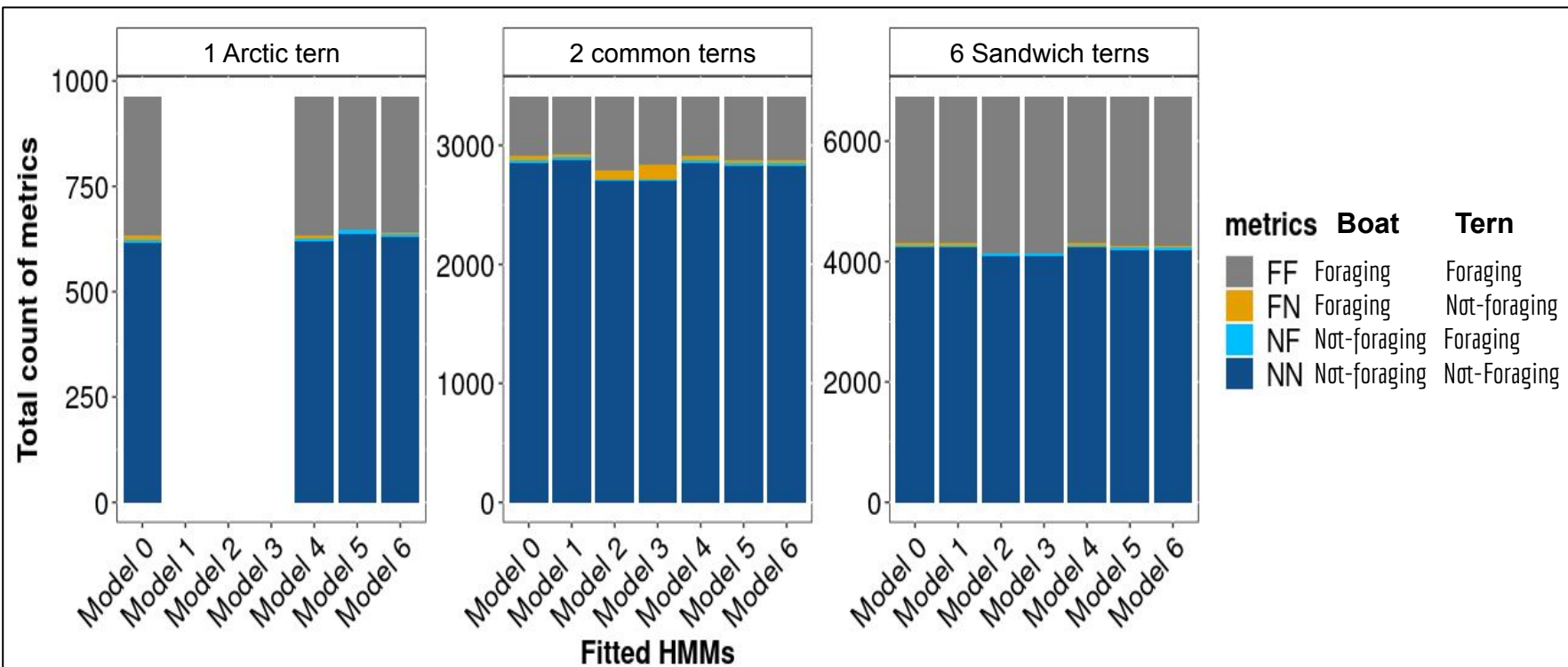


Figure: Confusion matrix metrics of inferred behavioural states from HMMs fitted to boat location and approximate location data in Coquet colony during chick-rearing, 2009. FF = true positives, FN = false negatives, NF = false positives, NN = true negatives

Hidden Markov Model (HMM)

An HMM is a time series model with an **observed** component driven by an underlying **hidden** component

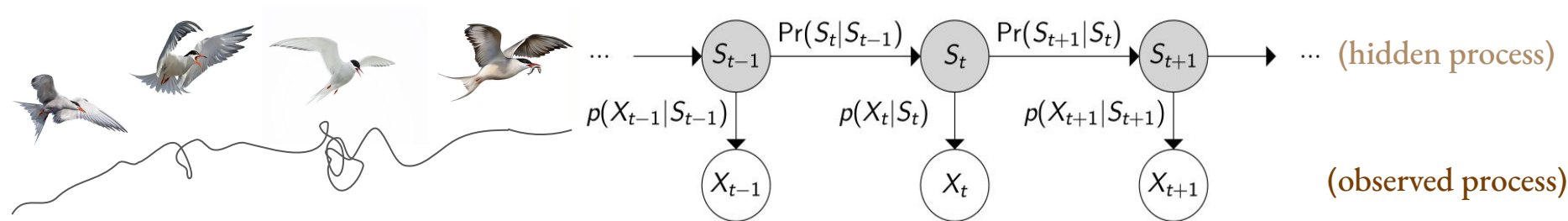


Figure: Representation of a HMM

- Foraging and Not-foraging behavioural states
- Metrics of step length and turning angles from boat GPS coordinates

Fitted Models

TABLE 1 HMMs (Models 0 - 6) fitted to the boat tracking data. Covariate = distance of the boat to the colony.

Models	Pooling effect		Covariate effect	
	State process	Observed process	State process	Observed process
0	✓	✓	✗	✗
1	✗	✓	✗	✗
2	✓	✗	✗	✗
3	✗	✗	✗	✗
4	✗	✓	✓	✗
5	✓	✗	✗	✓
6	✗	✗	✓	✓

Results

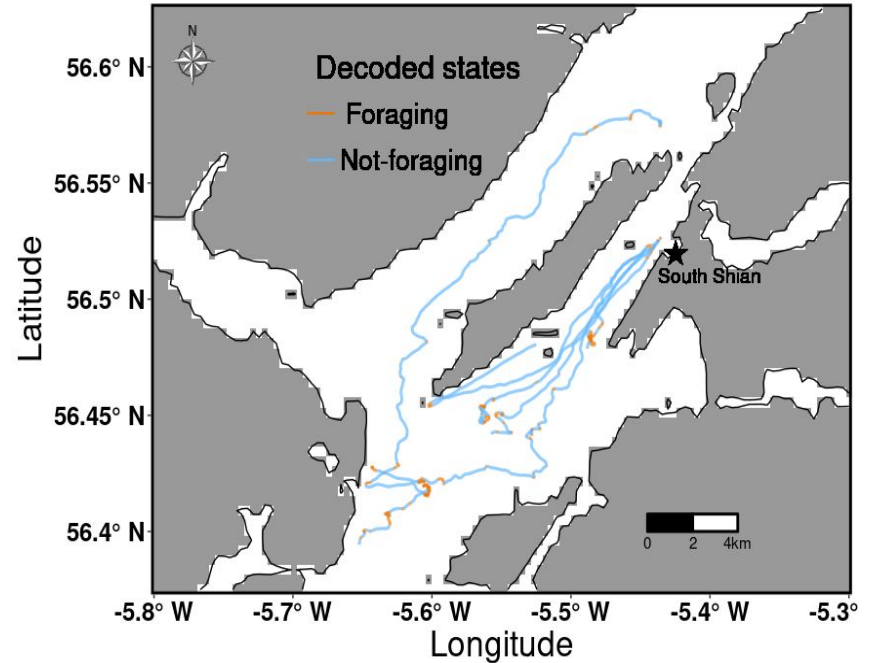
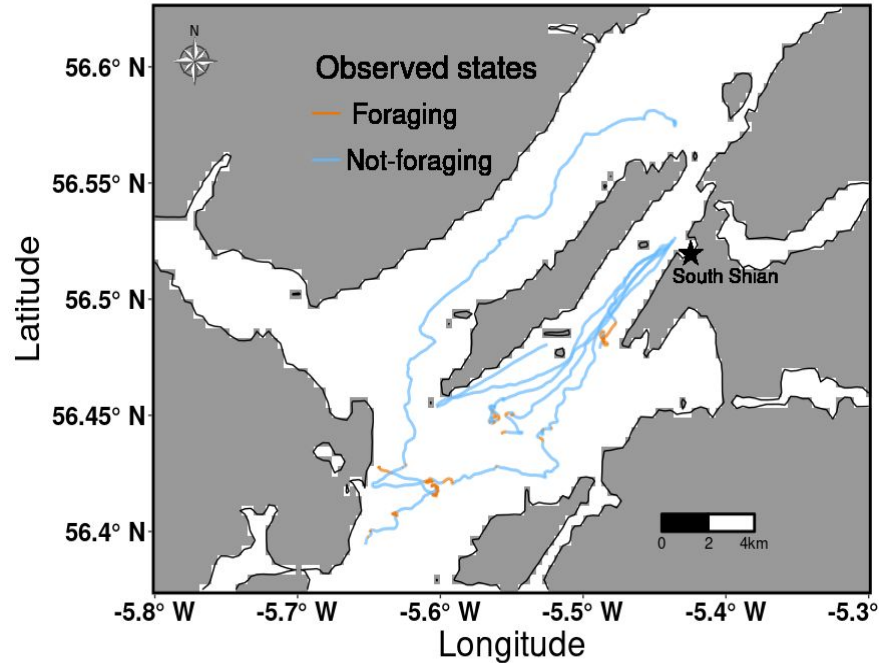


Figure 7: proxy (boat) movement data of 6 visually tracked common tern coloured with (a) observed and (b) decoded behavioural states: **foraging** and **not-foraging**, in **South Shian colony** () Mull, Scotland, during chick-rearing period, 2011.

Results

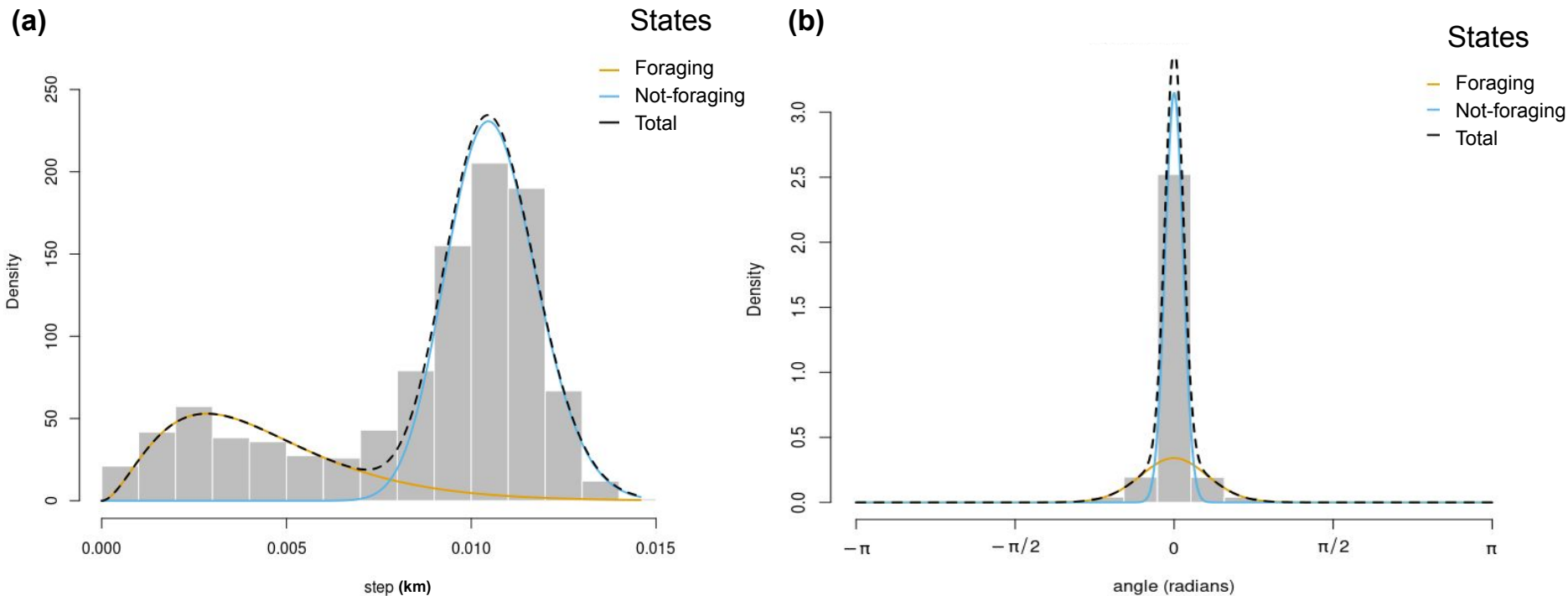


Figure : Histograms of the **(a)** proxy step length and **(b)** turning angle distributions of 6 visually-tracked common tern in **South Shian colony** during chick-rearing period, 2011. Lines represent the HMM fitted state-dependent distributions, and are coloured according to decoded behavioural state: **foraging** and **not-foraging**

Validation Result : incubation

- Decoded foraging behavioural state (true positive rate)



Arctic tern

69%

1 colony - Cockle



Common tern

60%

1 colony - Leith



Roseate tern

61%

1 colony - Blue Circle

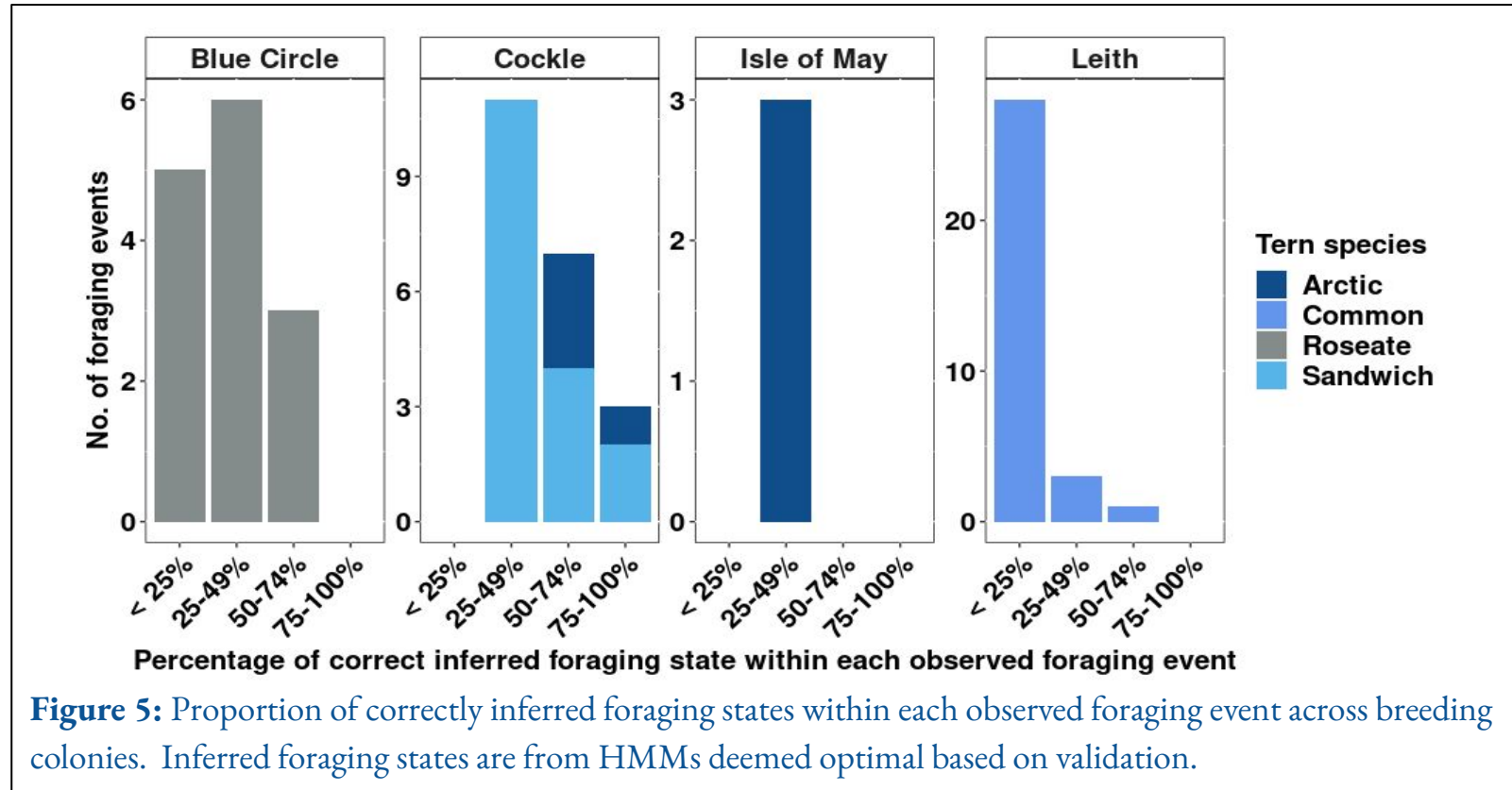


Sandwich tern

50%

1 colony - Cockle

Foraging event : incubation



Validation Result : chick-rearing

- Decoded foraging behavioural state (true positive rate)



Arctic tern

58-70%

3 colonies



Common tern

70-90%

5 colonies



Roseate tern

86%

1 colony



Sandwich tern

74-91%

4 colonies

Foraging event : chick-rearing

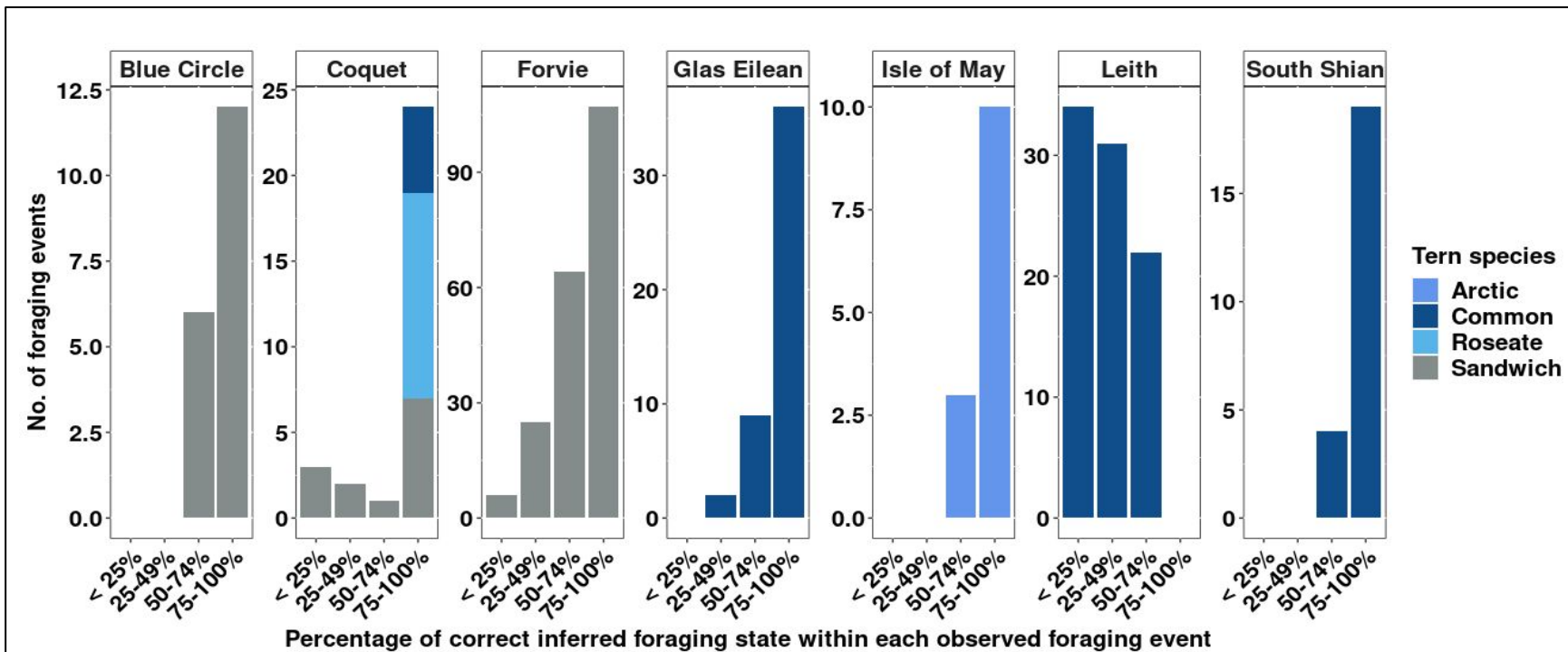
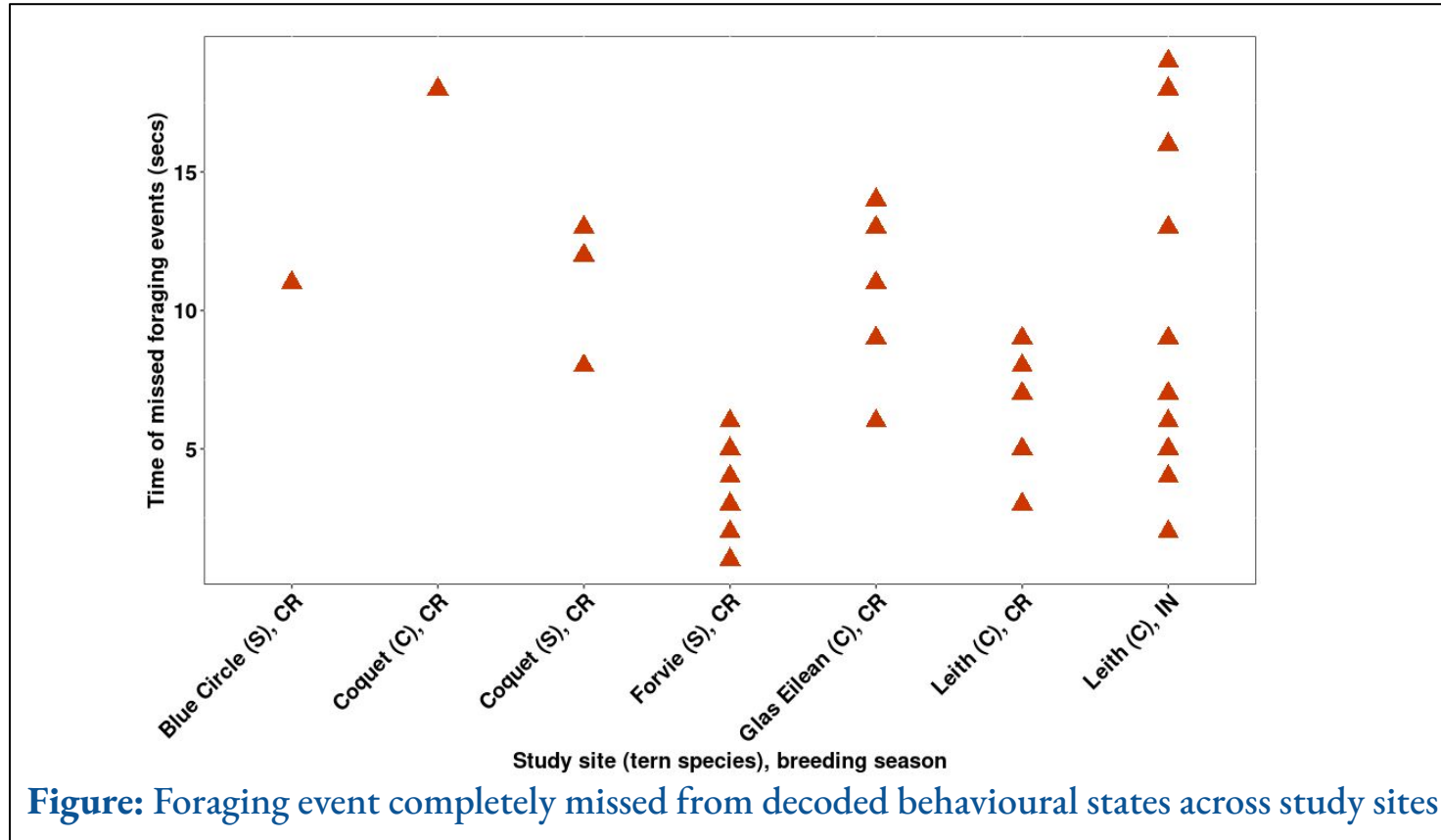


Figure 4: Proportion of correctly inferred foraging states captured within each observed foraging event across breeding colonies. Inferred foraging states are from HMMs deemed optimal based on validation.

Missed foraging event



Summary of findings



- Visual tracking data is suitable for our study as boat tracks closely replicate those from the approximate tern tracks
- HMMs failed to capture only few foraging bouts within the tracking data
- Important conservation-relevant behaviours inferred using HMMs is justified
- Distance of boat to the colony used as covariate in the model is not sufficient to account for the hidden behavioural state (foraging) from telemetry data during incubation period

Implication of findings



- Increased confidence in behaviours of seabirds inferred from telemetry data that are not validated
- Important conservation-relevant behaviours inferred using HMMs can inform the planning and design of designated protected areas
- Introducing relevant environmental covariates in HMMs in addition to the effect of distance of seabirds to the colony, may be helpful for decoding foraging behavioural states more accurately from telemetry during incubation season.

References

- Zucchini, W., MacDonald, I. L. & Langrock, R. (2016). **Hidden Markov Models for Time Series: An Introduction Using R** (2nd ed.)
- McClintock, B. T. & Michelot, T. (2018). **momentuhmm: R package for generalized hidden Markov models of animal movement**, *Methods in Ecology and Evolution* 9 (6), 1518–1530
- Perrow, M. R., Skeate, E. R. & Gilroy, J. J. (2011). **Visual tracking from a rigid-hulled inflatable boat to determine foraging movements of breeding terns**, *Journal of Field Ornithology* 82 (1), 68–79

Thanks for listening



Questions?